

# Measuring Neural-Network Robustness Under Adversarial Perturbation

## Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

Evidence status: Refreshed against current code snapshot 8412e98d38f28c2e4b43cdcbf8269133b346ca16 and successful CI run 36783579501 on 2026-09-30. The real CIFAR-10 robustness values are from the committed measured artifact results/cifar10\_smallcnn\_real.json; the current CI run validates the code/test surface but did not rerun the expensive CIFAR-10 benchmark job.

### Contribution

A reproducible robustness-evaluation harness for FGSM, PGD, and C&W that shows a measured 71.82% clean CIFAR-10 SmallCNN collapsing to 0.00% robust accuracy under PGD-20 at epsilon 8/255, while explicitly separating measured results from literature projections.

### Method

1. Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.
2. Measure clean accuracy on the full CIFAR-10 test set.
3. Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.
4. Measure robust accuracy on the committed attack subset.
5. Emit structured evidence and CI robustness-gate logic.

### Current Verified Evidence

Current-main Python 3.12 CI reports:

- 109 tests passed.
- 32.16% statement coverage; CI gate is 15%.
- Lint/format, security audit, and CodeQL jobs succeeded.
- The current CI robustness benchmark job is skipped on ordinary pushes, so the benchmark values below come from the committed real artifact rather than this CI run.

Committed measured CIFAR-10 artifact:

| Attack | Setting                    | Clean accuracy | Robust accuracy | Samples |
|--------|----------------------------|----------------|-----------------|---------|
| FGSM   | epsilon 8/255              | 71.82%         | 3.32%           | 1,024   |
| PGD-20 | epsilon 8/255, alpha 2/255 | 71.82%         | 0.00%           | 1,024   |
| C&W L2 | c=1.0, 100 steps           | 71.82%         | 4.20%           | 1,024   |

The model is a 1,117,354-parameter SmallCNN, trained for 6 epochs on CPU. Clean accuracy is measured on all 10,000 CIFAR-10 test examples.

### Honest Boundaries

- This repository measures vulnerability; it does not make a model robust.
- Robust-accuracy attacks use a 1,024-sample subset, not the full 10,000 test set.
- No AutoAttack, black-box transfer, or certified-robustness result is established.
- results/cifar10\_resnet18\_benchmark.json is explicitly literature-projected/synthetic and must not be presented as measured.

### Reproducibility

Expected current CI evidence: 109 passed, 32.16% coverage.